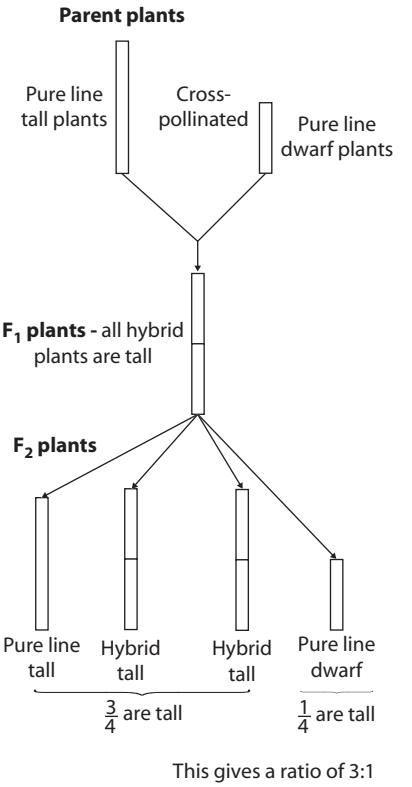
Mendel's pea plants (1)



This gives a ratio of 3:1

Gregor Mendel bred pea plants to study the laws of **heredity**. Peas grow quickly and there are many types. Mendel chose a pure line of tall peas to cross pollinate with a pure line of dwarf peas. He called the seeds from this pairing the F_1 (first generation) plants, or the F_1 generation. They were called **hybrids** because they grew from a cross between two genetically different plants. All the F_1 plants were tall.

- Mendel said that tallness **dominated** 'dwarfness'.
- Mendel called the tall gene **T** and the dwarf gene **t**.

Next, Mendel pollinated all the F_1 plants with one another. He collected their seeds and planted them. The plants that grew were called the F_2 plants or F_2 generation. Some F_2 plants were dwarf but many were tall. From this, Mendel concluded that dwarfness was caused by a **recessive** gene.

Punnet squares are drawn up to predict the **phenotype** of individuals from a genetic cross.

- The **genotype** refers to the individual's genes.
- The phenotype refers to the individual's appearance.

Below is a punnet square which shows the **genotypes** of the peas in the F_1 cross.

	T	**t**
T	TT pure line tall plants	Tt hybrid tall plants
t	tT hybrid tall plants	tt pure line dwarf plants

¼ TT
¼ Tt = ¾ tall plants
¼ tT

¼ tt = ¼ dwarf plants

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Mendel's pea plants (2)

☞ **1** Use the word bank to complete the passage below.

Word bank
Mendel dwarf tall plant represented dominated seeds

When M_____ cross-pollinated tall and d_____ plants, he planted their s_____ to produce the F_1 generation. Every p_____ from the F_1 generation was t_____. Mendel declared that tallness had d_____ the **recessive** dwarfness. He r_____ tallness as 'T' and dwarfness as 't'.

2 Order the steps Mendel took to study the laws of heredity.

	Step number
Plant F_1 seeds and watch them grow into F_2 plants	
Cross-pollinate tall and dwarf plants (parents)	
Plant parent seeds and watch them grow into F_1 plants	
Collect seeds of parent plants	
Cross-pollinate all F_1 plants then collect their seeds	

3 What is the phenotype of a pea plant with the genotype **Tt**? _____

4 What is the phenotype of a pea plant with the genotype **TT**? _____

5 What is the phenotype of a pea plant with the genotype **tt**? _____

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